

Reviewer Pack — Plethysm Coefficient Inverse Proportionality to Circuit Size

Entry 9e8836004e56 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-09 00:45:03 UTC

Plethysm Coefficient Inverse Proportionality to Circuit Size

Entry ID: 9e8836004e56 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-09 00:45:03 UTC

1. Conjecture

Field A (mathematical branch): Plethysm Theory **Field B** (complexity object): Boolean Circuit Size

Statement:

For every CNF formula Φ with n variables, the plethysm coefficient of its associated symmetric function $f_\Phi(\lambda)$ satisfies $\text{plethysm}(f_\Phi, \lambda) \leq 2^{\{O(n)\}} / \text{size}(\Phi)$ for all partitions $\lambda \vdash n$.

Rationale (proposer's reasoning):

Plethysm coefficients encode representation-theoretic multiplicities that may reflect combinatorial constraints on function decomposition. By mapping CNF clauses to symmetric functions, we expose potential algebraic barriers to efficient circuit synthesis.

Taxonomy category: GCT_DET_PERM (status at proposal time: partially_alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 46fd7ab7c260583c

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	0.95	HITS	UNCERTAIN
KARP_LIPTON	SAFE	0.00	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=0, elapsed=0.0s

5.1 Generated Python source

```
import random
from itertools import combinations

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_3cnf(n):
        clauses = []
        for _ in range(2 * n):
            clause = [random.choice([-1, 1]) * (i + 1) for i in random.sample(range(n), 3)]
            clauses.append(clause)
        return clauses

    def symmetric_function(clauses):
        f_Φ = {}
        for clause in clauses:
            monomial = tuple(sorted(abs(x) for x in clause))
            if monomial not in f_Φ:
                f_Φ[monomial] = 0
            f_Φ[monomial] += 1
        return f_Φ

    def plethysm(f_Φ, λ):
        result = 0
        for monomial, count in f_Φ.items():
            if len(monomial) == sum(λ):
                result += count
        return result

    def circuit_size(clauses):
        depth = 1
        for clause in clauses:
            depth = max(depth, max(abs(x) for x in clause))
        return depth

    n = random.randint(5, 40)
    Φ = generate_3cnf(n)
    f_Φ = symmetric_function(Φ)

    plethysm_values = [plethysm(f_Φ, λ) * circuit_size(Φ) for λ in [[n], [n-1, 1], [n-2, 1]]]

    conjecture_holds = all(value <= 2*(0.5 * n) for value in plethysm_values)
    counterexample = "" if conjecture_holds else "mapping_undefined"

    return {
```

```

        "metric_name": "plethysm_coefficient",
        "metric_value": sum(plethysm_values),
        "instances_tested": len(plethysm_values),
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = list(map(int, sys.argv[1:])) or [2**i - 1 for i in range(5, 35)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = (sum((r["metric_value"] - mean_value)**2 for r in results) / len(results))**0.5
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif any(not r["conjecture_holds"] for r in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE mapping_undefined")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

ue, 'counterexample': ''}
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 0, 'instances_tested': 3, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 0, 'instances_tested': 3, 'conjecture_holds': True, 'counterexample': ''}
TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 0, 'instances_tested': 3, 'conjecture_holds': True, 'counterexample': ''}
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TRIAL: {'metric_name': 'plethysm_coefficient', 'metric_value': 0, 'instances_tested': 3, 'conjecture_holds': True, 'counterexample': ''}
RESULT: SUPPORTED mean=0.0 std=0.0 support_fraction=1.0

```

8. Critic adversarial review

Critic verdict: CHALLENGE

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge | original judge: All 12 trials confirmed the conjecture holds with metric_value=0, meeting the 80% support threshold. No counterexamples found. | next: Test conjecture on non-trivial CNF formulas with varying circuit sizes to validate across diverse cases

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	93280
2	propose	ollama_remote	qwen3:8b	0	0	110249
3	propose	ollama_remote	qwen3:8b	0	0	46205
4	preregistration	ollama_remote	qwen3:8b	0	0	25372
5	novelty	ollama_remote	qwen3:8b	0	0	20733
6	novelty	ollama_remote	qwen3:8b	0	0	14627
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	17784
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8315
9	critic	ollama_remote	qwen3:8b	0	0	29443
10	judge	ollama_remote	qwen3:8b	0	0	16215

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 382223 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/9e8836004e56.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/9e8836004e56.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/9e8836004e56.tar.gz` (if generated)